Языковое моделирование;
Работа с последовательностями

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Эмбеддинги для разных языков и косинусная мера близости

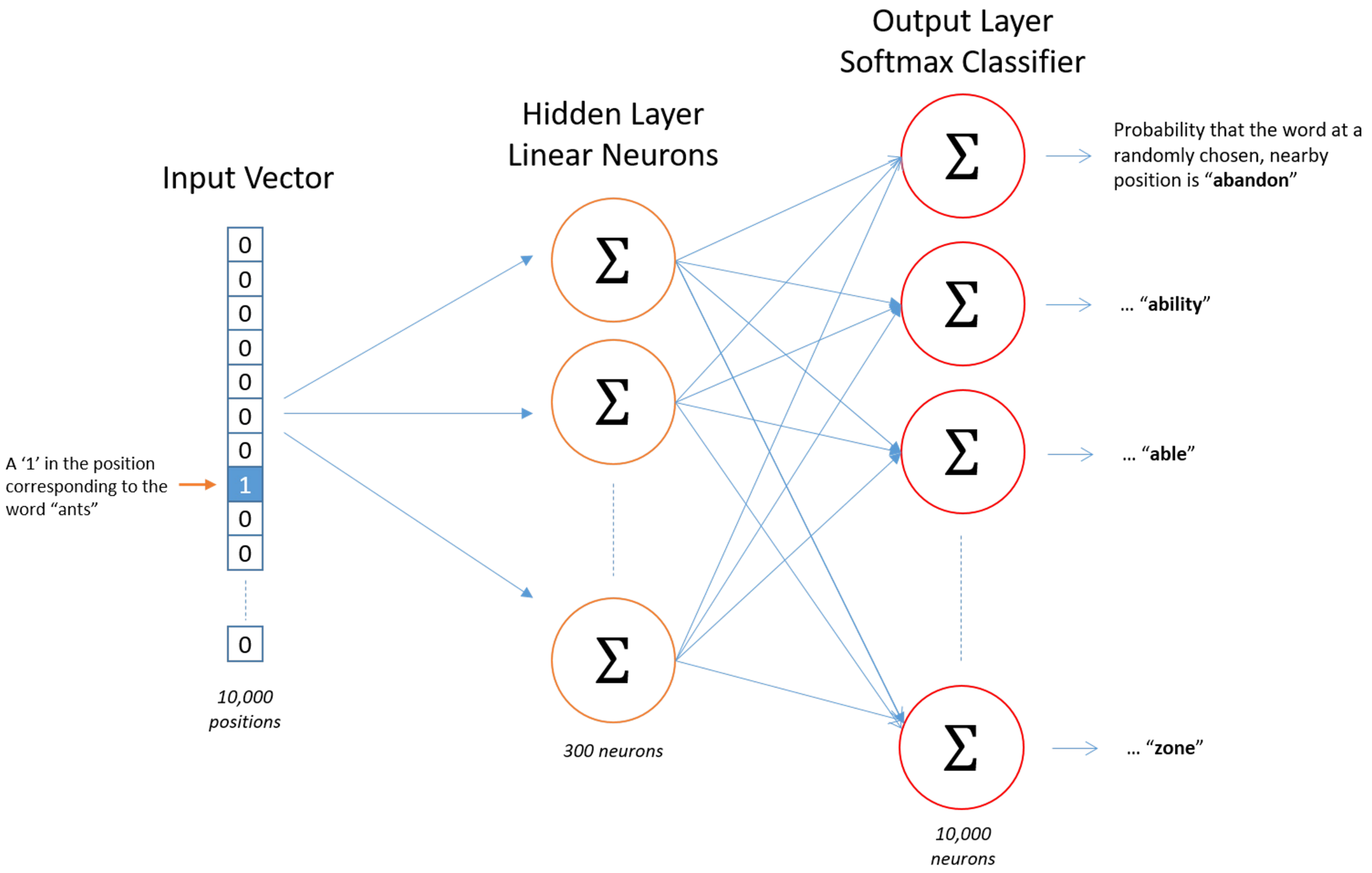
01

Embeddings: word2vec

Source Text		Training Samples
The quick brown fox jumps over the lazy dog.	→	(the, quick) (the, brown)
The **quick** brown fox jumps over the lazy dog.	→	(quick, the) (quick, brown) (quick, fox)
The quick **brown** fox jumps over the lazy dog.	→	(brown, the) (brown, quick) (brown, fox) (brown, jumps)
The quick brown **fox** jumps over the lazy dog.	→	(fox, quick) (fox, brown) (fox, jumps) (fox, over)

Source: http://mccormickml.com/2016/04/19/word2vec-tutorial-the-skip-gram-model/

Embeddings: word2vec



Source: http://mccormickml.com/2016/04/19/word2vec-tutorial-the-skip-gram-model/

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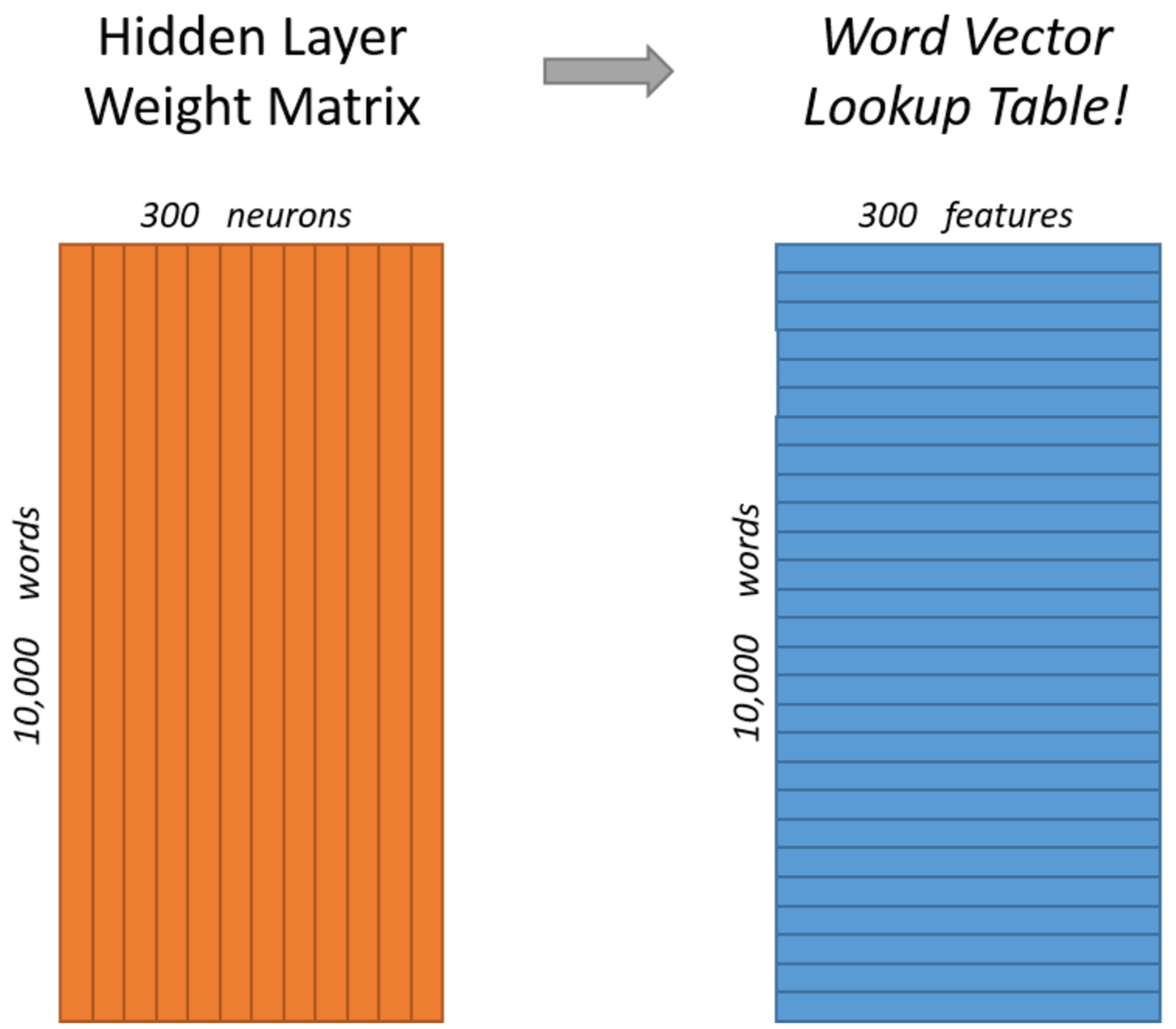
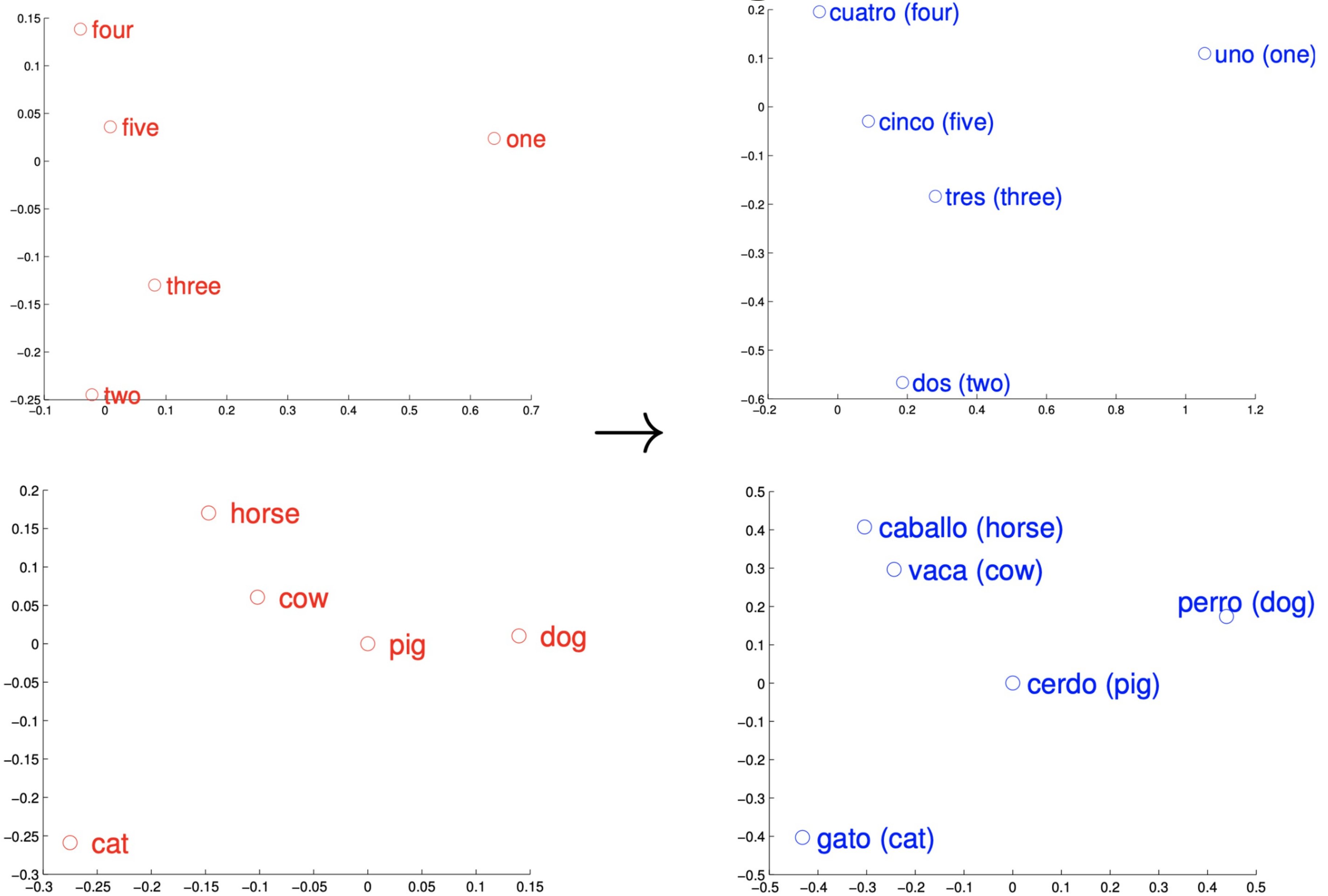


Source: http://mccormickml.com/2016/04/19/word2vec-tutorial-the-skip-gram-model/

Word embeddings in different languages



Source: Exploiting similarities among languages for machine translation, 2013

Word embeddings in different languages

- Word embeddings are quite similar for different languages
- Assume there n = 5000 word-translation pairs $\{x_i, y_i\}_{i \in \{1,n\}}$
- Learn linear mapping between the source and target spaces

$$W^\star = \underset{W \in M_d(\mathbb{R})}{\operatorname{argmin}} \|WX - Y\|_\mathrm{F}$$

- The translation of source word is $t = \operatorname{argmax}_t \cos(Wx_s, y_t)$

Sources: [Exploiting similarities among languages for machine translation, 2013]
[Word translation without parallel data, ICLR 2018]

Word embeddings in different languages

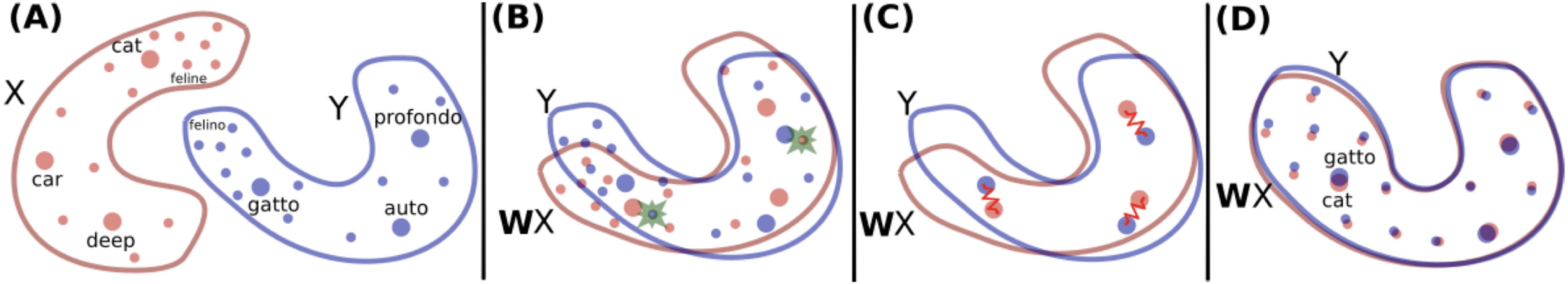
- Word embeddings are quite similar for different languages
- Assume there n = 5000 word-translation pairs $\{x_i, y_i\}_{i \in \{1,n\}}$
- Learn linear mapping between the source and target spaces **enforcing an orthogonality constraint** on : W

$$W^\star = \underset{W \in O_d(\mathbb{R})}{\operatorname{argmin}} \|WX - Y\|_{\mathrm{F}} = UV^T, \text{with } U\Sigma V^T = \mathrm{SVD}(YX^T).$$

- The translation of source word is $t = \operatorname{argmax}_t \cos(Wx_s, y_t)$

Sources: [Normalized word embedding and orthogonal transform for bilingual word translation, NAACL 2015](#)
[Word translation without parallel data, ICLR 2018](#)

Word embeddings in different languages



Comment: mapping between two languages can be done completely in unsupervised manner with GANs.

We will meet them later.

More info available in the mentioned paper:

Source: [Word translation without parallel data, ICLR 2018]

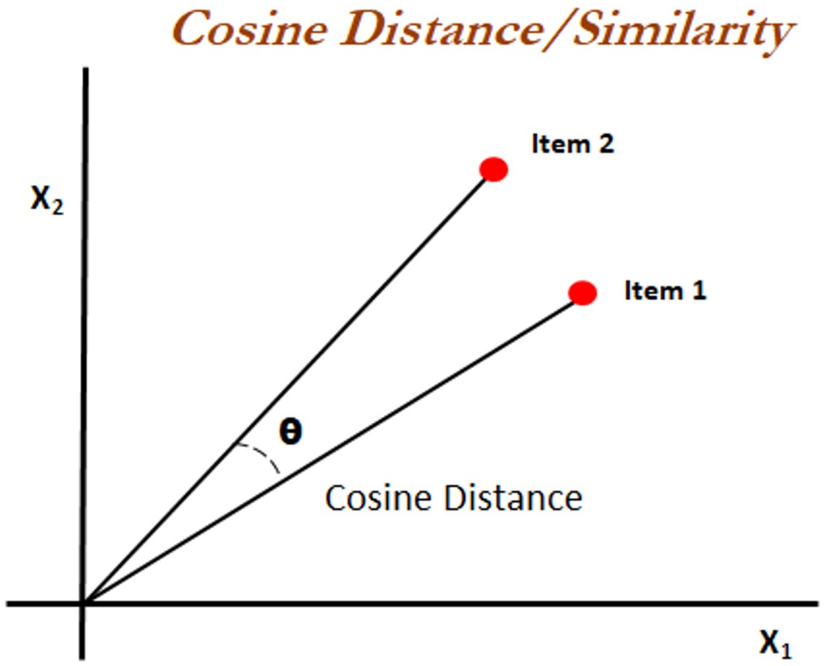
Why cosine distance/similarity?

$$\text{similarity} = \cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\|\|\mathbf{B}\|} = \frac{\sum_{i=1}^{n} A_i B_i}{\sqrt{\sum_{i=1}^{n} A_i^2} \sqrt{\sum_{i=1}^{n} B_i^2}}$$

Cosine distance focuses on angle between the vectors.

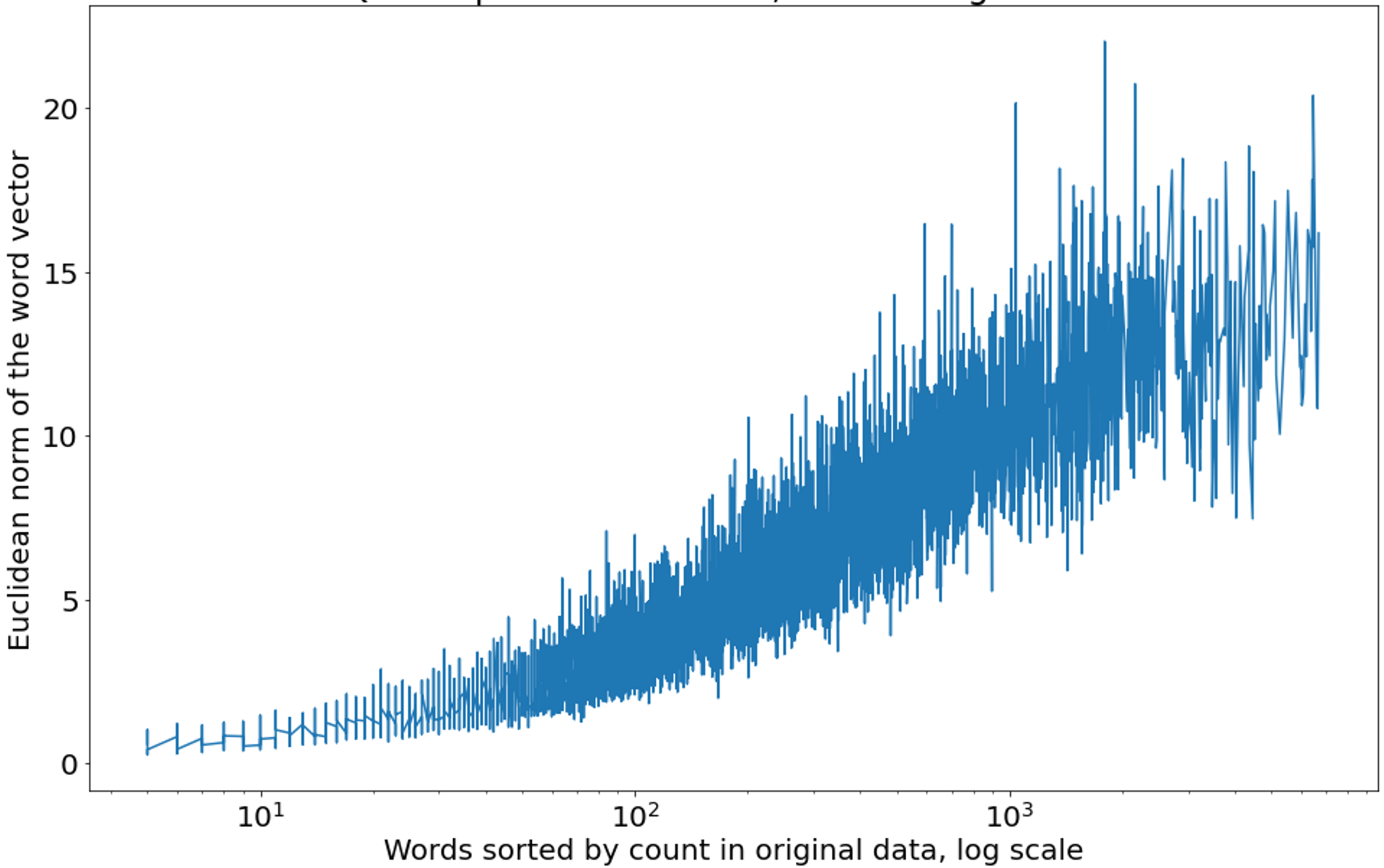
With count-based approaches (e.g. BOW) it is really useful.

With word embeddings it is useful as well.



Source: question

How word frequency affects the embedding vector norm
Quora questions dataset, embedding size 32



Vector norms for words with no specific context

word	count	vector norm
overheat	11	0.81233
enormous	12	0.807057
dog	1212	11.2591
cat	1545	10.3738
laptop	1906	14.5192
phone	4124	15.7901
a	155726	11.4656
the	252068	8.47355

Генеративные модели до ChatGPT

02

Давным-давно, в далёком-далёком 2012

Shakespeare

```
PANDARUS:
Alas, I think he shall be come approached and the day
When little srain would be attain'd into being never fed,
And who is but a chain and subjects of his death,
I should not sleep.

Second Senator:
They are away this miseries, produced upon my soul,
Breaking and strongly should be buried, when I perish
The earth and thoughts of many states.

DUKE VINCENTIO:
Well, your wit is in the care of side and that.

Second Lord:
They would be ruled after this chamber, and
my fair nues begun out of the fact, to be conveyed,
Whose noble souls I'll have the heart of the wars.

Clown:
Come, sir, I will make did behold your worship.

VIOLA:
I'll drink it.
```

Algebraic Geometry (Latex)

Proof. Omitted. □

Lemma 0.1. *Let $\mathcal{C}$ be a set of the construction.*

Let $\mathcal{C}$ be a gerber covering. Let $\mathcal{F}$ be a quasi-coherent sheaves of $\mathcal{O}$-modules. We have to show that

$$\mathcal{O}_{\mathcal{O}_X} = \mathcal{O}_X(\mathcal{L})$$

.

Proof. This is an algebraic space with the composition of sheaves $\mathcal{F}$ on $X_{étale}$ we have

$$\mathcal{O}_X(\mathcal{F}) = \{morph_1 \times_{\mathcal{O}_X} (\mathcal{G}, \mathcal{F})\}$$

where $\mathcal{G}$ defines an isomorphism $\mathcal{F} \to \mathcal{F}$ of $\mathcal{O}$-modules. □

Lemma 0.2. *This is an integer $\mathcal{Z}$ is injective.*

Proof. See Spaces, Lemma ??. □

Lemma 0.3. *Let S be a scheme. Let X be a scheme and X is an affine open covering. Let $\mathcal{U} \subset \mathcal{X}$ be a canonical and locally of finite type. Let X be a scheme. Let X be a scheme which is equal to the formal complex.*

The following to the construction of the lemma follows.

Let X be a scheme. Let X be a scheme covering. Let

$$b : X \to Y' \to Y \to Y \to Y' \times_X Y \to X.$$

be a morphism of algebraic spaces over S and Y.

Proof. Let X be a nonzero scheme of X. Let X be an algebraic space. Let $\mathcal{F}$ be a quasi-coherent sheaf of $\mathcal{O}_X$-modules. The following are equivalent

(1) $\mathcal{F}$ is an algebraic space over S.
(2) If X is an affine open covering.

Consider a common structure on X and X the functor $\mathcal{O}_X(U)$ which is locally of finite type. □

Linux kernel (source code)

```c
/*
 * If this error is set, we will need anything right after that BSD.
 */
static void action_new_function(struct s_stat_info *wb)
{
  unsigned long flags;
  int lel_idx_bit = e->edd, *sys & ~((unsigned long) *FIRST_COMPAT);
  buf[0] = 0xFFFFFFFF & (bit << 4);
  min(inc, slist->bytes);
  printk(KERN_WARNING "Memory allocated %02x/%02x, "
    "original MLL instead\n"),
    min(min(multi_run - s->len, max) * num_data_in),
    frame_pos, sz + first_seg);
  div_u64_w(val, inb_p);
  spin_unlock(&disk->queue_lock);
  mutex_unlock(&s->sock->mutex);
  mutex_unlock(&func->mutex);
  return disassemble(info->pending_bh);
}
```


source: http://karpathy.github.io/2015/05/21/rnn-effectiveness/

Давным-давно, в далёком-далёком 2012

Proof. Omitted. □

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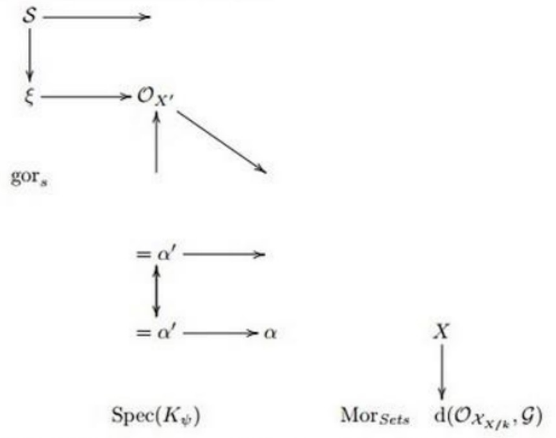
This since $\mathcal{F} \in \mathcal{F}$ and $x \in \mathcal{G}$ the diagram



is a limit. Then $\mathcal{G}$ is a finite type and assume S is a flat and $\mathcal{F}$ and $\mathcal{G}$ is a finite type f_*. This is of finite type diagrams, and

- the composition of $\mathcal{G}$ is a regular sequence,
- $\mathcal{O}_{X'}$ is a sheaf of rings.

□

Proof. We have see that $X = \mathrm{Spec}(R)$ and $\mathcal{F}$ is a finite type representable by algebraic space. The property $\mathcal{F}$ is a finite morphism of algebraic stacks. Then the cohomology of X is an open neighbourhood of U. □

Proof. This is clear that $\mathcal{G}$ is a finite presentation, see Lemmas ??.
A *reduced above* we conclude that U is an open covering of $\mathcal{C}$. The functor $\mathcal{F}$ is a "field

$$\mathcal{O}_{X,x} \longrightarrow \mathcal{F}_{\overline{x}} \quad \text{-1}(\mathcal{O}_{X_{\acute{e}tale}}) \longrightarrow \mathcal{O}_{X_\ell}^{-1}\mathcal{O}_{X_\lambda}(\mathcal{O}_{X_\eta}^{\overline{v}})$$

is an isomorphism of covering of $\mathcal{O}_{X_i}$. If $\mathcal{F}$ is the unique element of $\mathcal{F}$ such that X is an isomorphism.
The property $\mathcal{F}$ is a disjoint union of Proposition ?? and we can filtered set of presentations of a scheme $\mathcal{O}_X$-algebra with $\mathcal{F}$ are opens of finite type over S.
If $\mathcal{F}$ is a scheme theoretic image points. □

If $\mathcal{F}$ is a finite direct sum $\mathcal{O}_{X_\lambda}$ is a closed immersion, see Lemma ??. This is a sequence of $\mathcal{F}$ is a similar morphism.

source: http://karpathy.github.io/2015/05/21/rnn-effectiveness/

Давным-давно, в далёком-далёком 2012

```
#include <asm/io.h>
#include <asm/prom.h>
#include <asm/e820.h>
#include <asm/system_info.h>
#include <asm/setew.h>
#include <asm/pgproto.h>

#define REG_PG    vesa_slot_addr_pack
#define PFM_NOCOMP  AFSR(0, load)
#define STACK_DDR(type)     (func)

#define SWAP_ALLOCATE(nr)     (e)
#define emulate_sigs()  arch_get_unaligned_child()
#define access_rw(TST)  asm volatile("movd %%esp, %0, %3" : : "r" (0));   \
  if (__type & DO_READ)

static void stat_PC_SEC __read_mostly offsetof(struct seq_argsqueue, \
          pC>[1]);

static void
os_prefix(unsigned long sys)
{
#ifdef CONFIG_PREEMPT
  PUT_PARAM_RAID(2, sel) = get_state_state();
  set_pid_sum((unsigned long)state, current_state_str(),
           (unsigned long)-1->lr_full; low;
}
```

source: http://karpathy.github.io/2015/05/21/rnn-effectiveness/

Обработка последовательностей

03

Последовательности формально

Последовательность объектов/событий: $x_1, x_2, \ldots, x_t$

Каждый объект – вектор: $x_i \in \mathbb{R}$

Задача: научиться обрабатывать последовательности переменной длины;

Последовательности всюду: тексты, видео, действия пользователей, действия агентов и пр.

Марковское свойство

Последовательность $x_1, x_2, \ldots, x_t$ обладает марковским свойством:

$$P(x_{t+1}|x_1, x_2, \ldots, x_t) = P(x_{t+1}|x_t)$$

Здесь мы предполагаем, что для каждого элемента задано вероятностное распределение (формально говоря, мы рассматриваем случайный процесс).

Т.е. в марковском процессе играет роль лишь фиксированная часть истории.

Как работать с последовательностью переменной длины?

Глобальные статистики

$$z = \frac{1}{T}\sum_{i=1}^{T} x_i$$

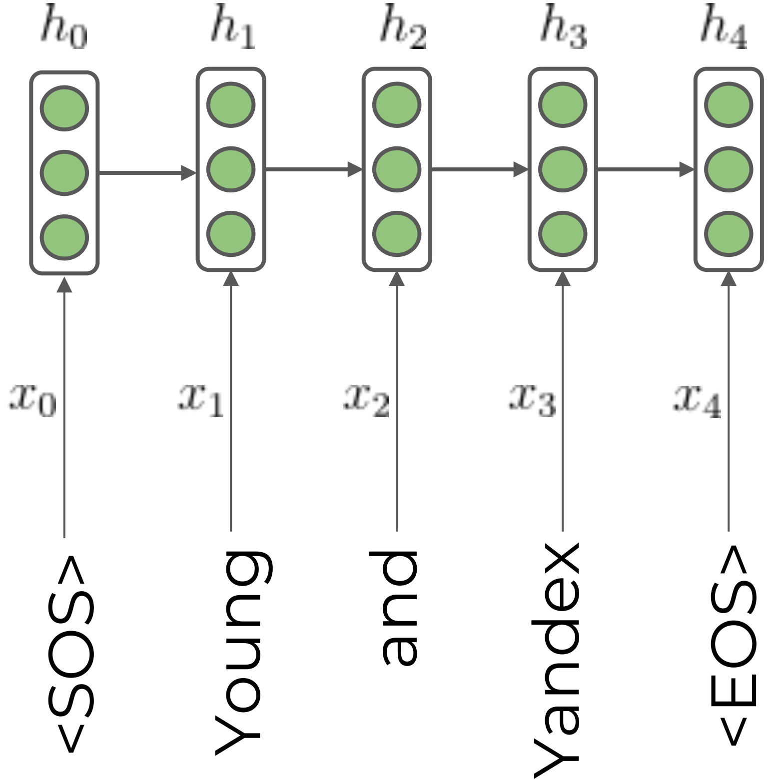
Пошаговая обработка

$$\boxed{h_{t+1} = f_\theta(h_t, x_t)}$$

Рекуррентный блок RNN

04

Кодирование последовательностей

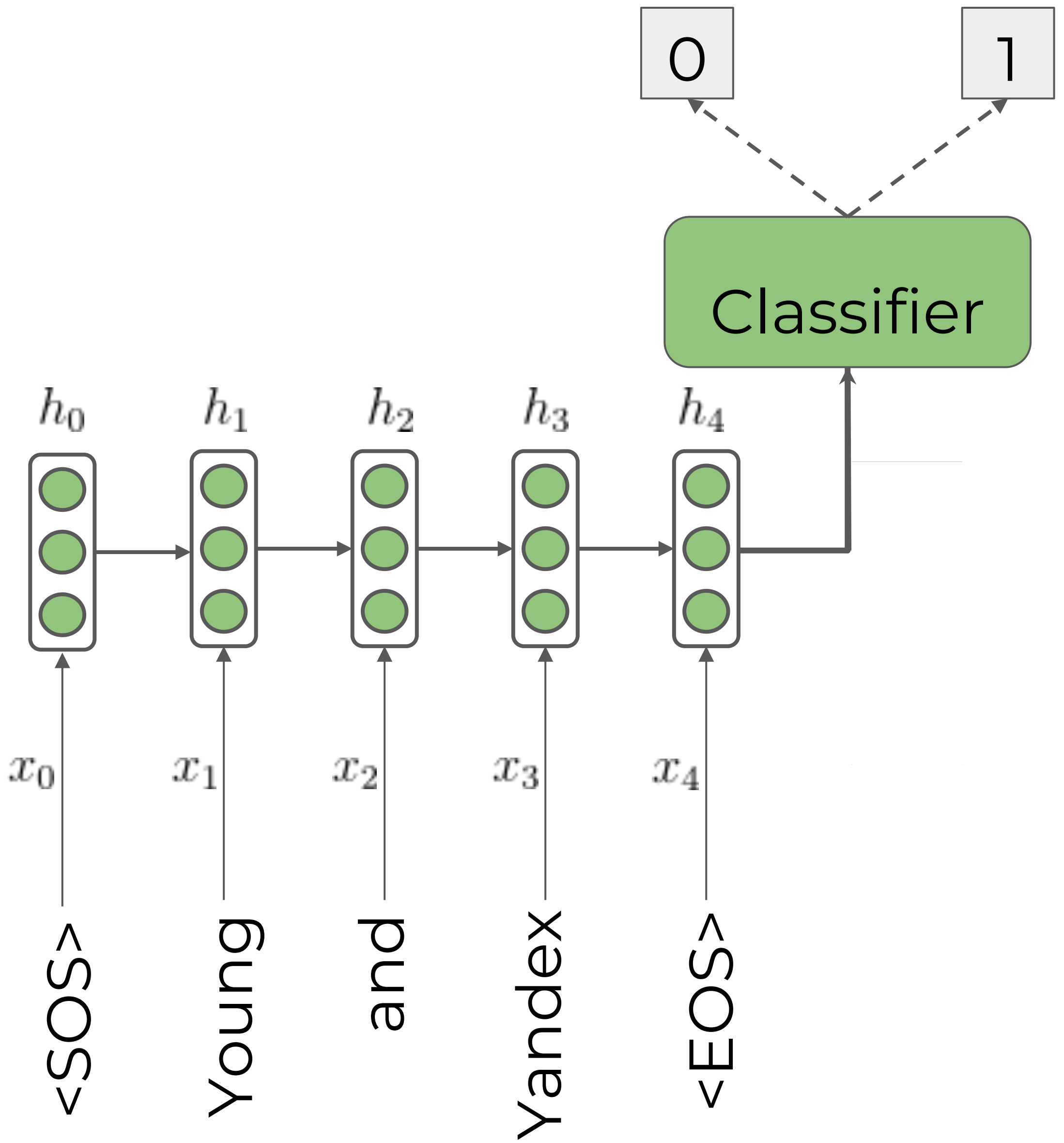


Если бы могли закодировать текущий контекст в виде вектора h_t и обогащать его новой информацией итеративным образом…

Что же за функция ниже?

$$f_\theta(h_t, x_t)$$

Классификация последовательностей

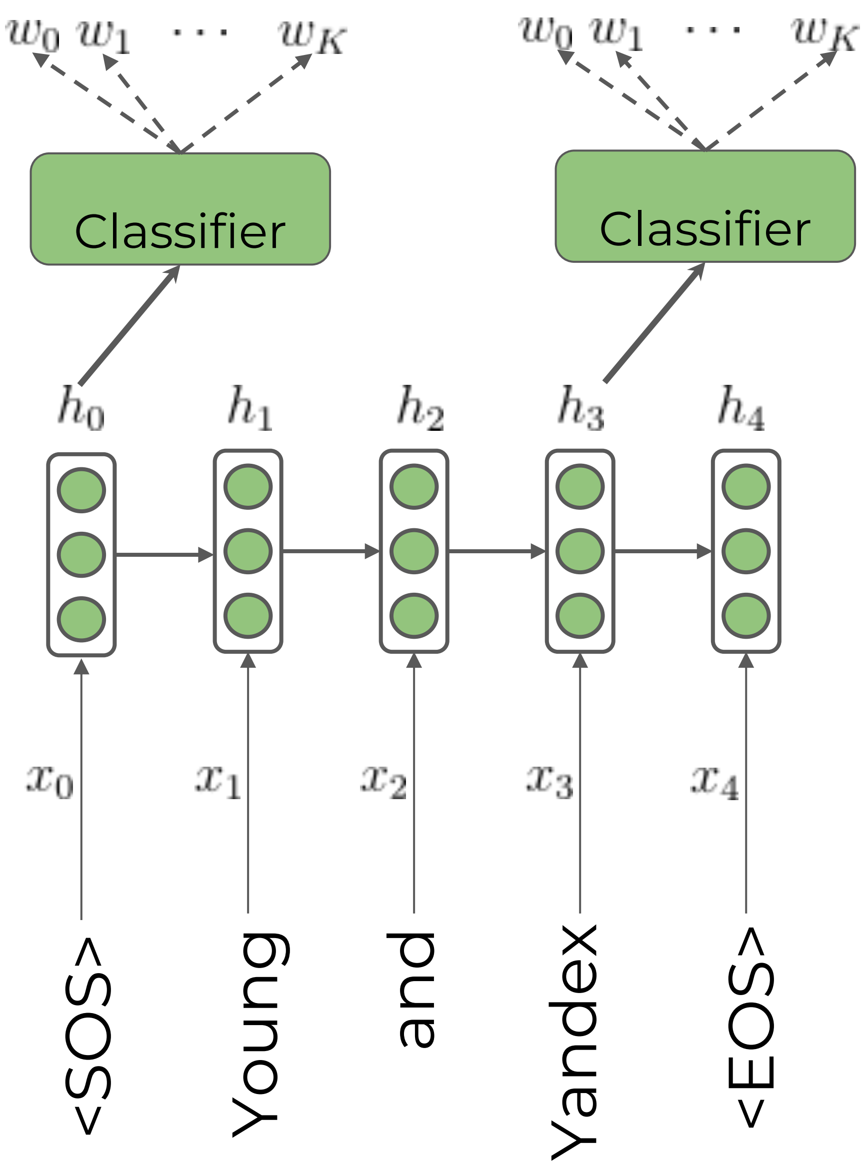


Тогда вектор h_t выступал бы эмбеддингом всего левого контекста.

Мы могли бы решать задачу классификации последовательностей.

$$\hat{y} \sim P_\phi(y|h_t)$$

Генерация последовательностей

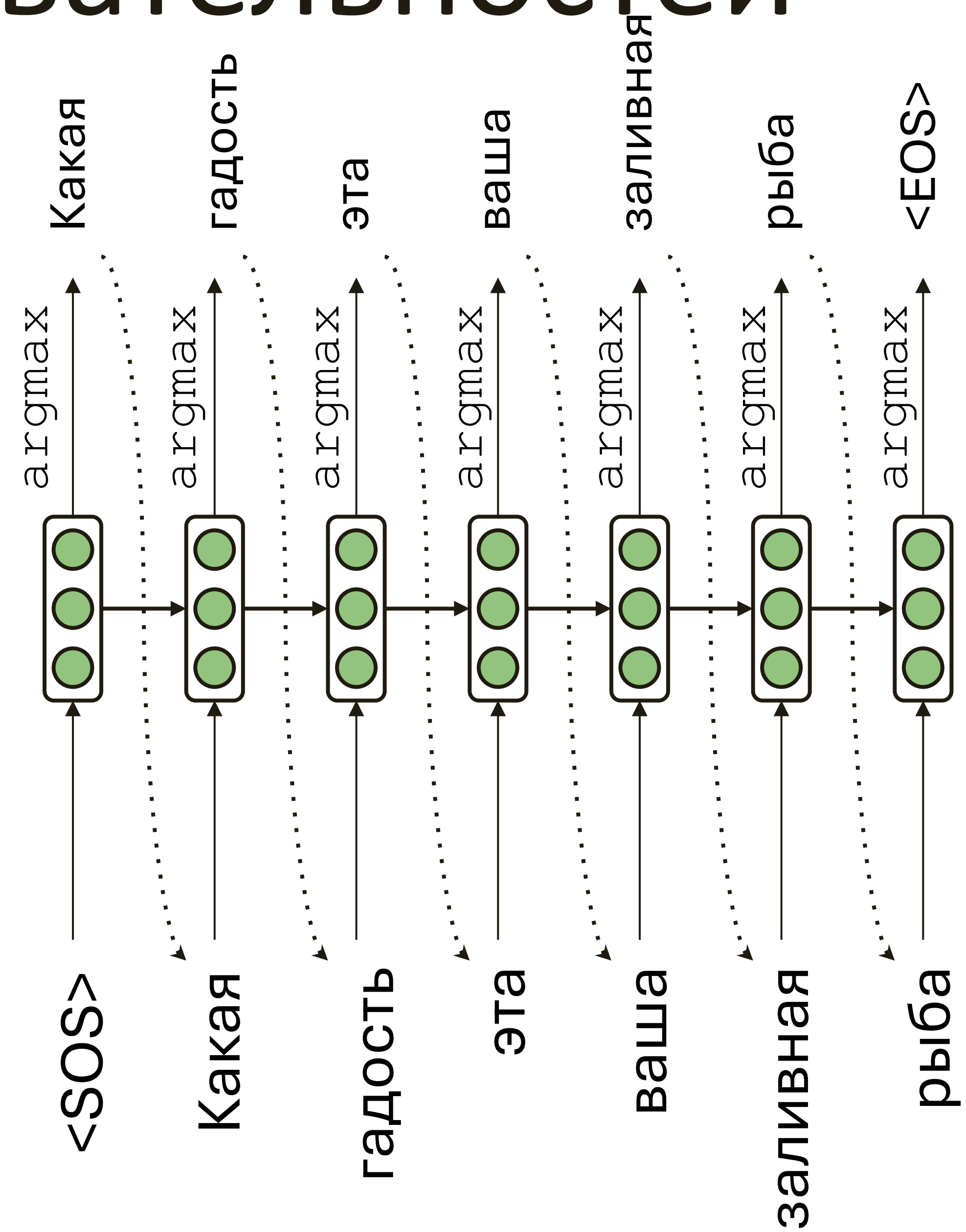


Тогда вектор h_t выступал бы эмбеддингом всего левого контекста.

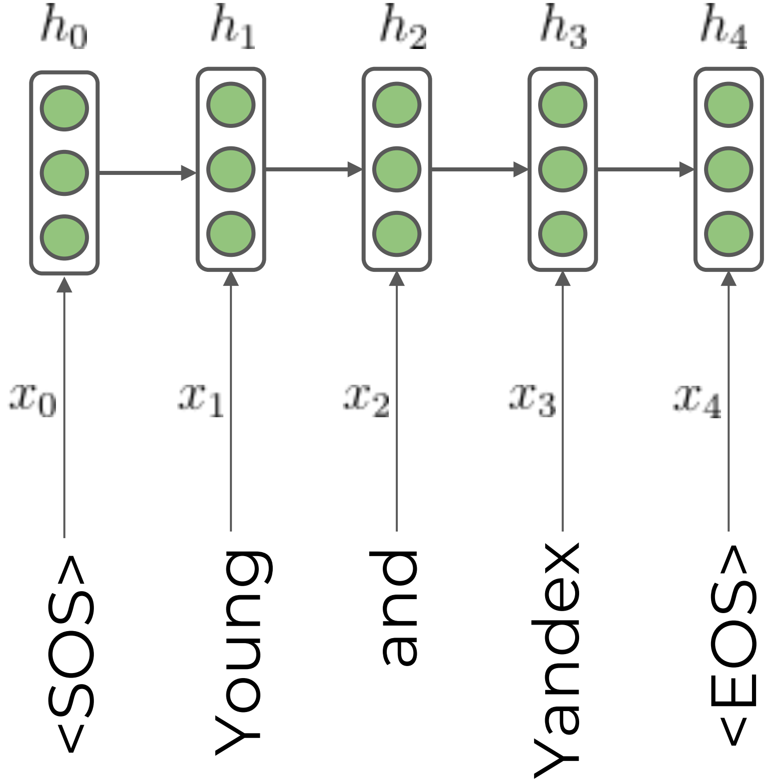
Мы могли бы решать задачу генерации последовательностей, предсказывая следующий токен на каждом шаге:

$$\hat{x}_{t+1} \sim P_\gamma(w|h_t)$$

Генерация последовательностей



Кодирование последовательностей



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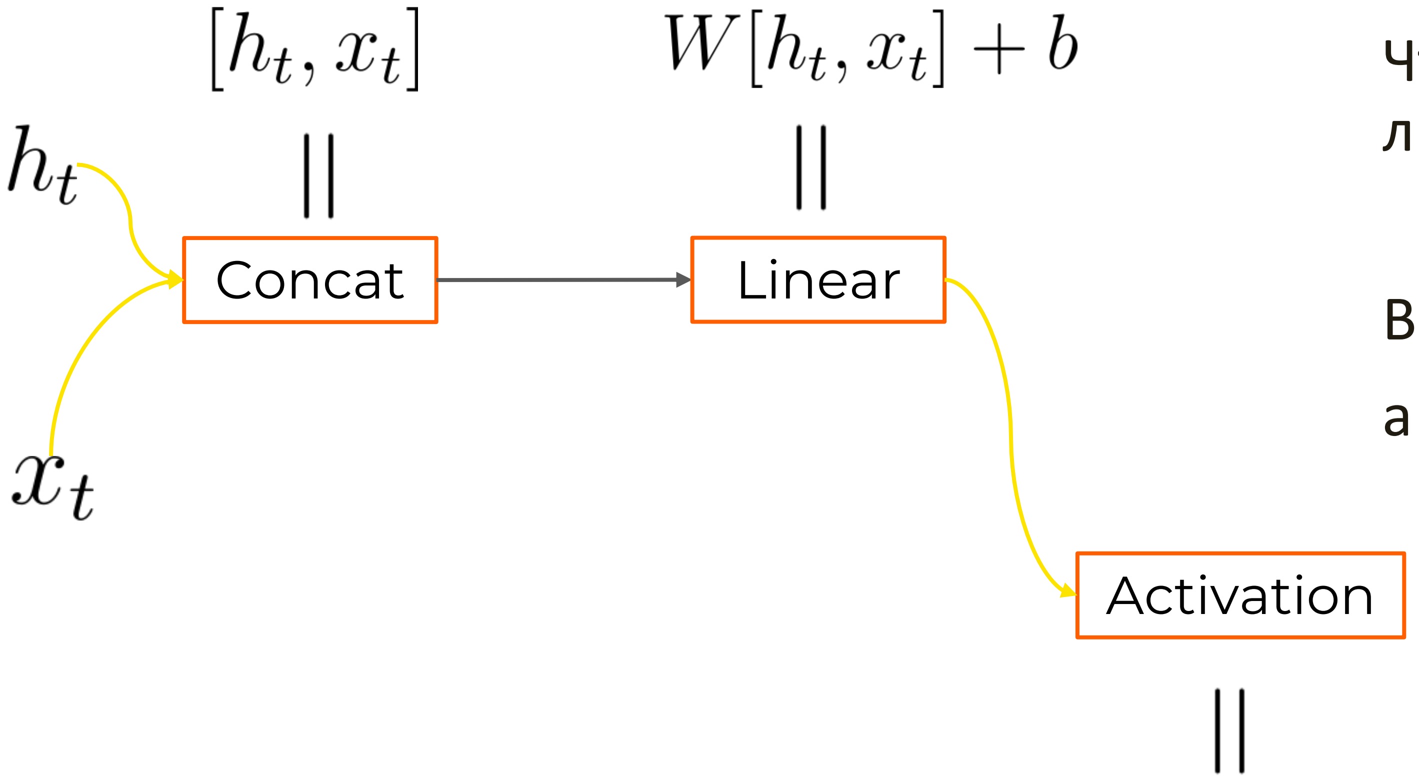
Что же за функция ниже?

$$f_\theta(h_t, x_t)$$

Рекурретный блок – основа RNN



Что может быть проще одного линейного слоя?

В данном случае $\theta = \{W, b\}$,

а [...] означает конкатенацию.

$$f_\theta(h_t, x_t) = \sigma(W[h_t, x_t] + b)$$

RNN в виде формул:

$$h_0 = \overline{0}$$

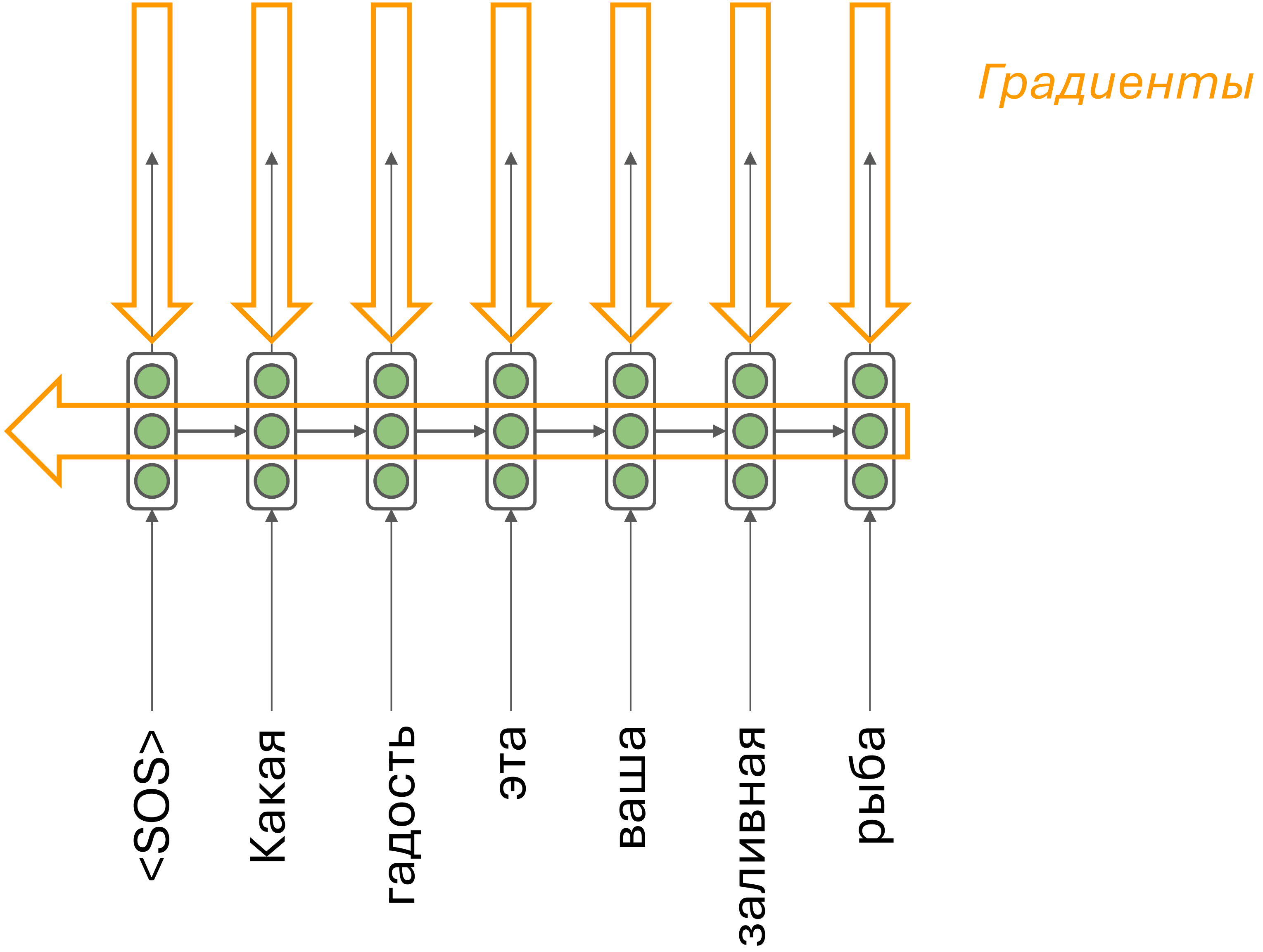
$$h_1 = \sigma\left(W[h_0, x_0] + b\right)$$

$$h_2 = \sigma\left(W[h_1, x_1] + b\right) = \sigma\left(W[\sigma\left(W_{\text{hid}}[h_0, x_0] + b, x_1\right)] + b\right)$$

$$h_{i+1} = \sigma\left(W_{\text{hid}}[h_i, x_i] + b\right)$$

$$x_{i+1} \sim P(w|h_i) = \text{softmax}\left(W_{\text{out}} h_i + b_{\text{out}}\right)$$

Как настраивать параметры?



Проблемы с градиентами

05

Взрыв градиента (exploding gradient)

Что если градиент окажется очень большим?

$$\theta^{new} = \theta^{old} - \overbrace{\alpha}^{\text{learning rate}} \underbrace{\nabla_\theta J(\theta)}_{\text{gradient}}$$

Оптимизационный процесс может просто разойтись!

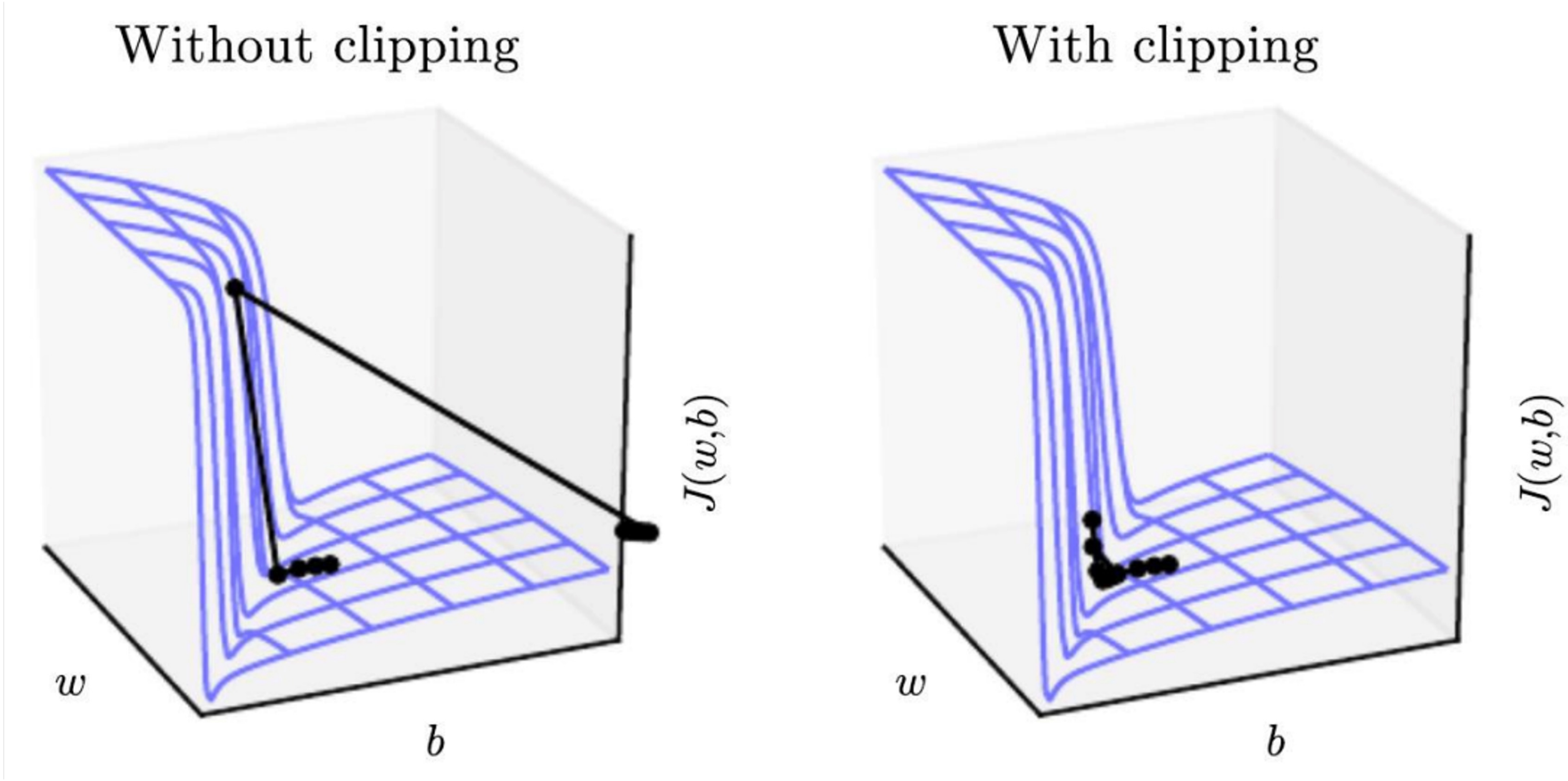
Взрыв градиента (exploding gradient)

Можно нормировать градиент, приводя к наибольшей допустимой норме:

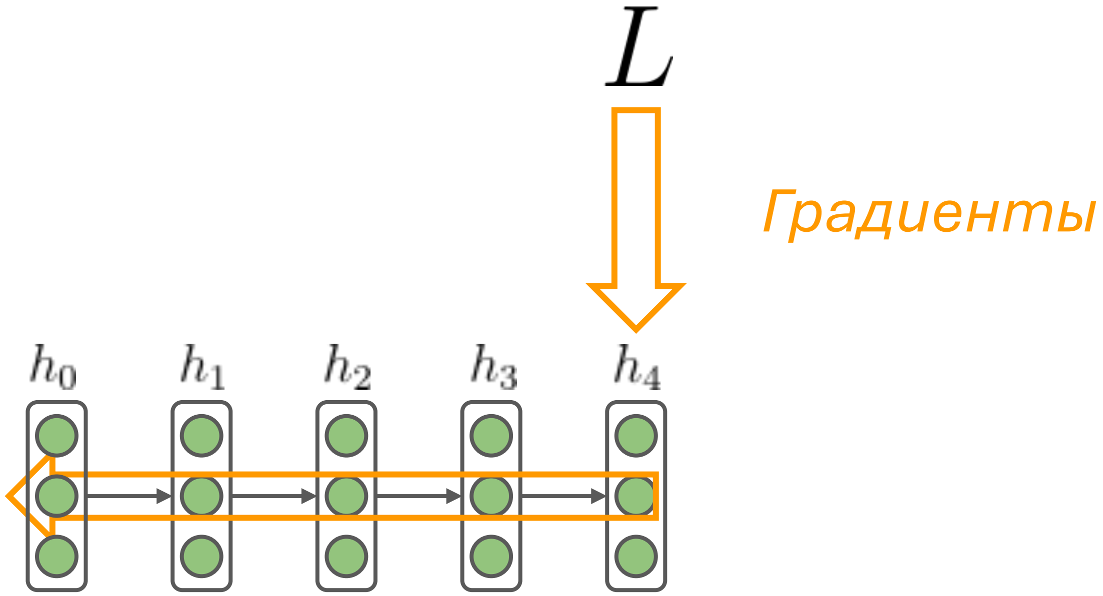
Algorithm 1 Pseudo-code for norm clipping

$\hat{\mathbf{g}} \leftarrow \frac{\partial \mathcal{E}}{\partial \theta}$
if $\|\hat{\mathbf{g}}\| \geq threshold$ **then**
  $\hat{\mathbf{g}} \leftarrow \frac{threshold}{\|\hat{\mathbf{g}}\|} \hat{\mathbf{g}}$
end if

Взрыв градиента (exploding gradient)



Затухающий градиент (vanishing gradient)

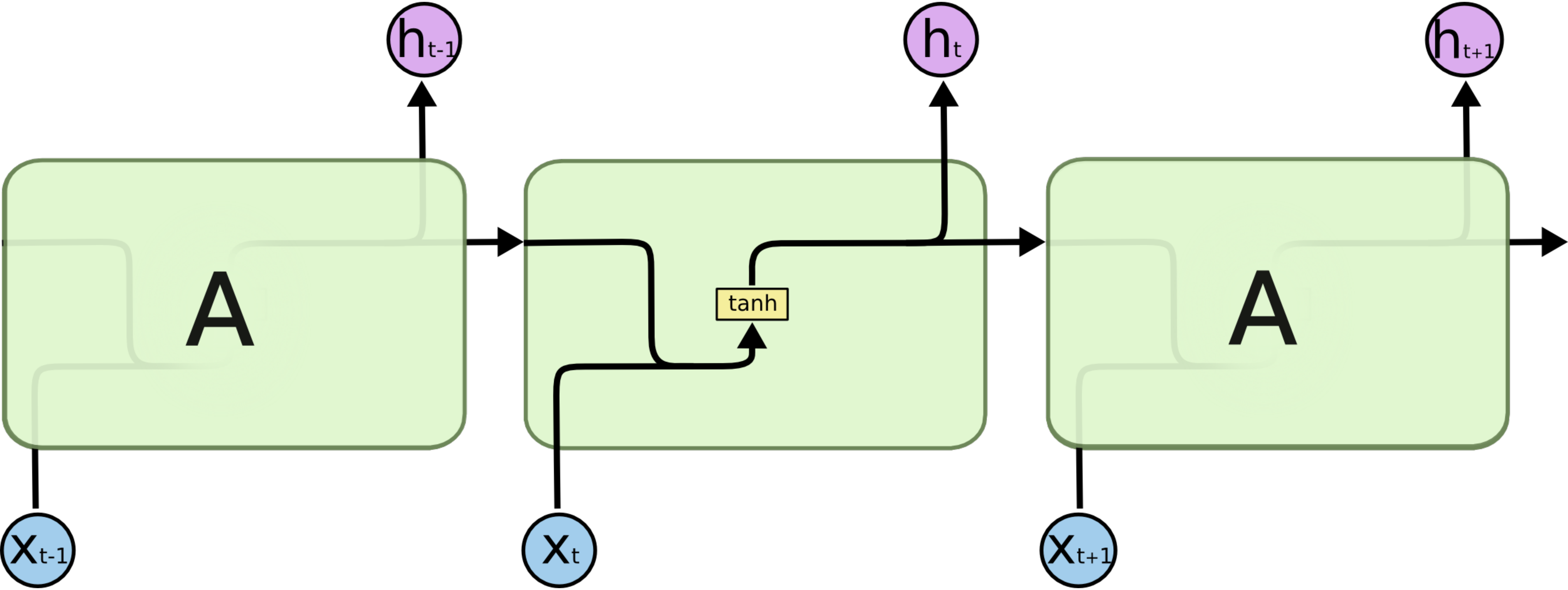


$$\frac{\partial L}{\partial h_0} = \frac{\partial L}{\partial h_4}\frac{\partial h_4}{\partial h_3}\frac{\partial h_3}{\partial h_2}\frac{\partial h_2}{\partial h_1}\frac{\partial h_1}{\partial h_0}$$

Усложняя RNN – LSTM (как пример)

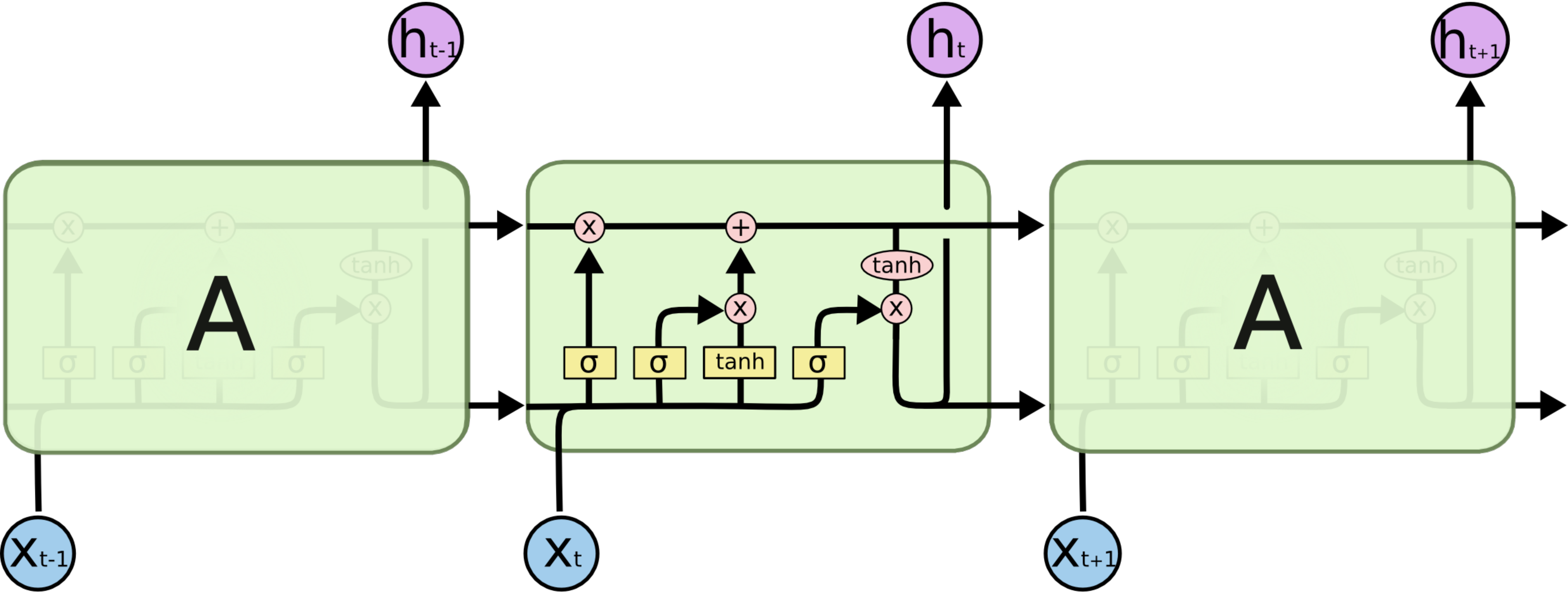
06

Vanilla RNN –> LSTM



source: https://colah.github.io/posts/2015-08-Understanding-LSTMs/

Vanilla RNN –> LSTM



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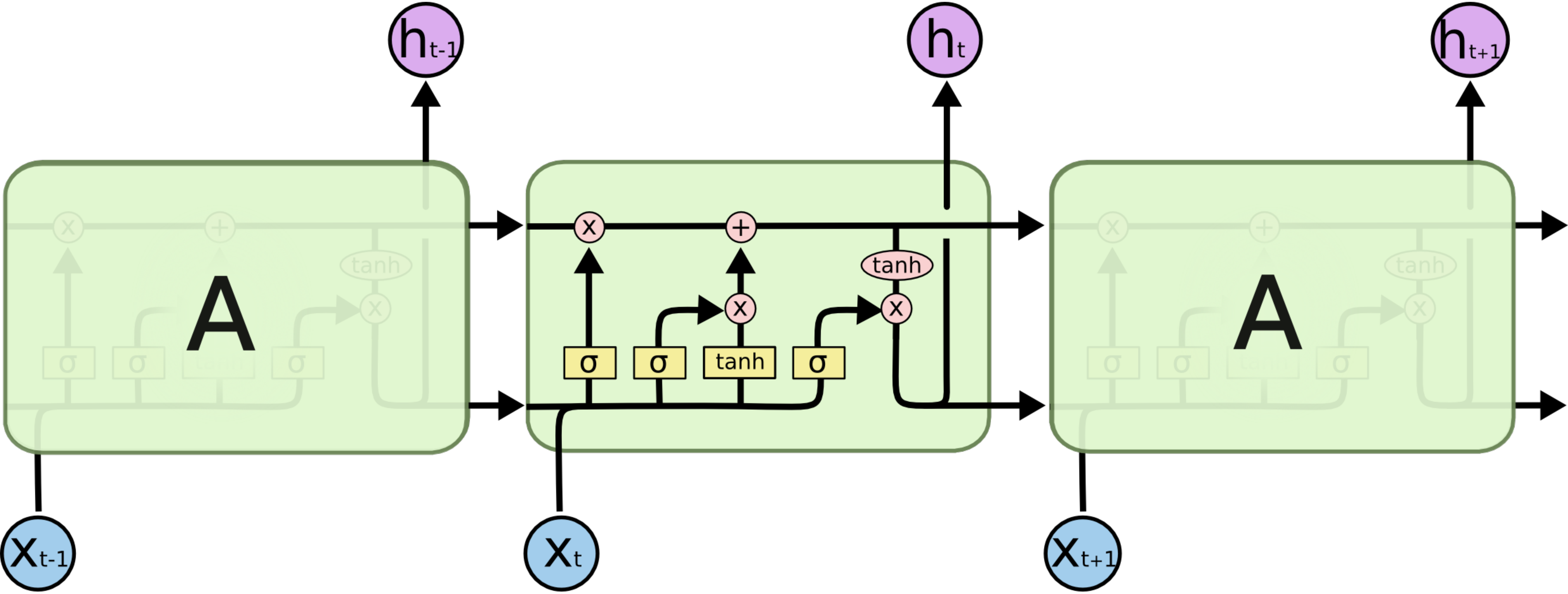
Vanilla RNN –> LSTM



source: https://colah.github.io/posts/2015-08-Understanding-LSTMs/

Благодарю за внимание

@Rads_ai

Канал
с заметками по
мотивам занятий и
много чем еще